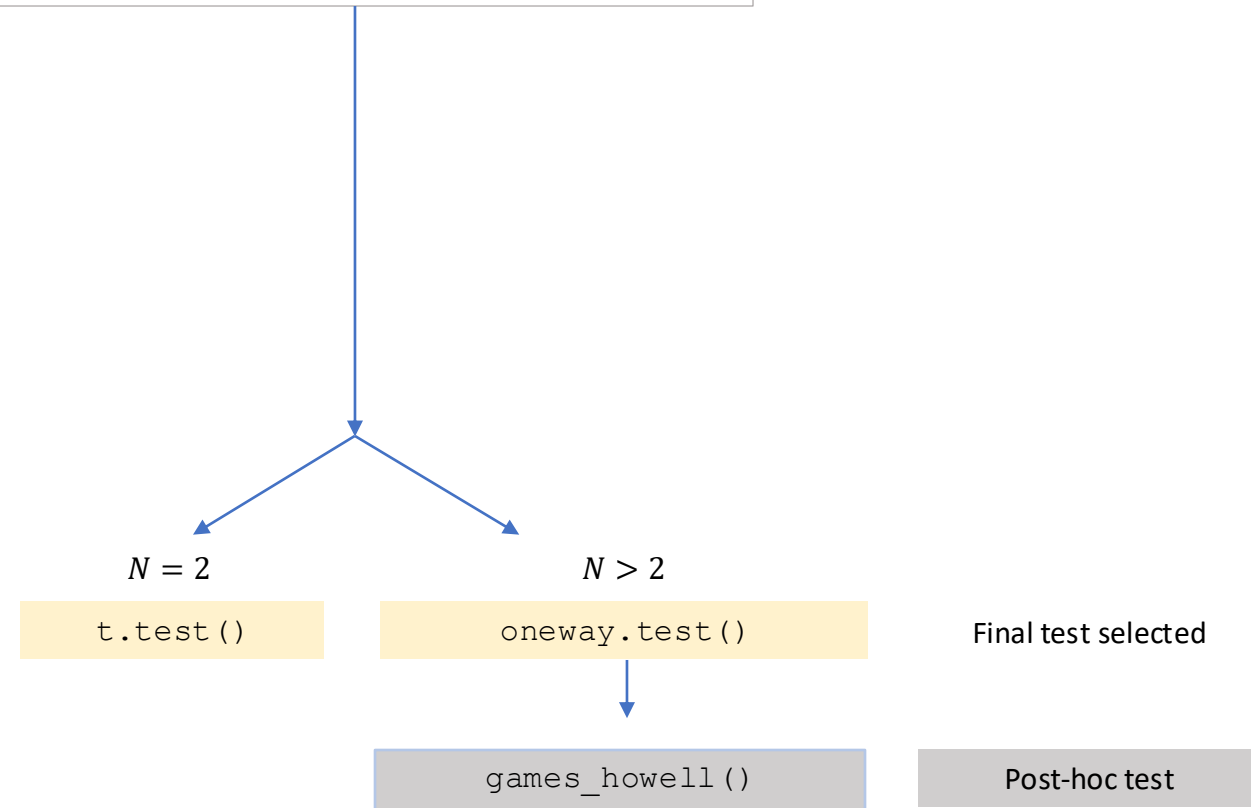
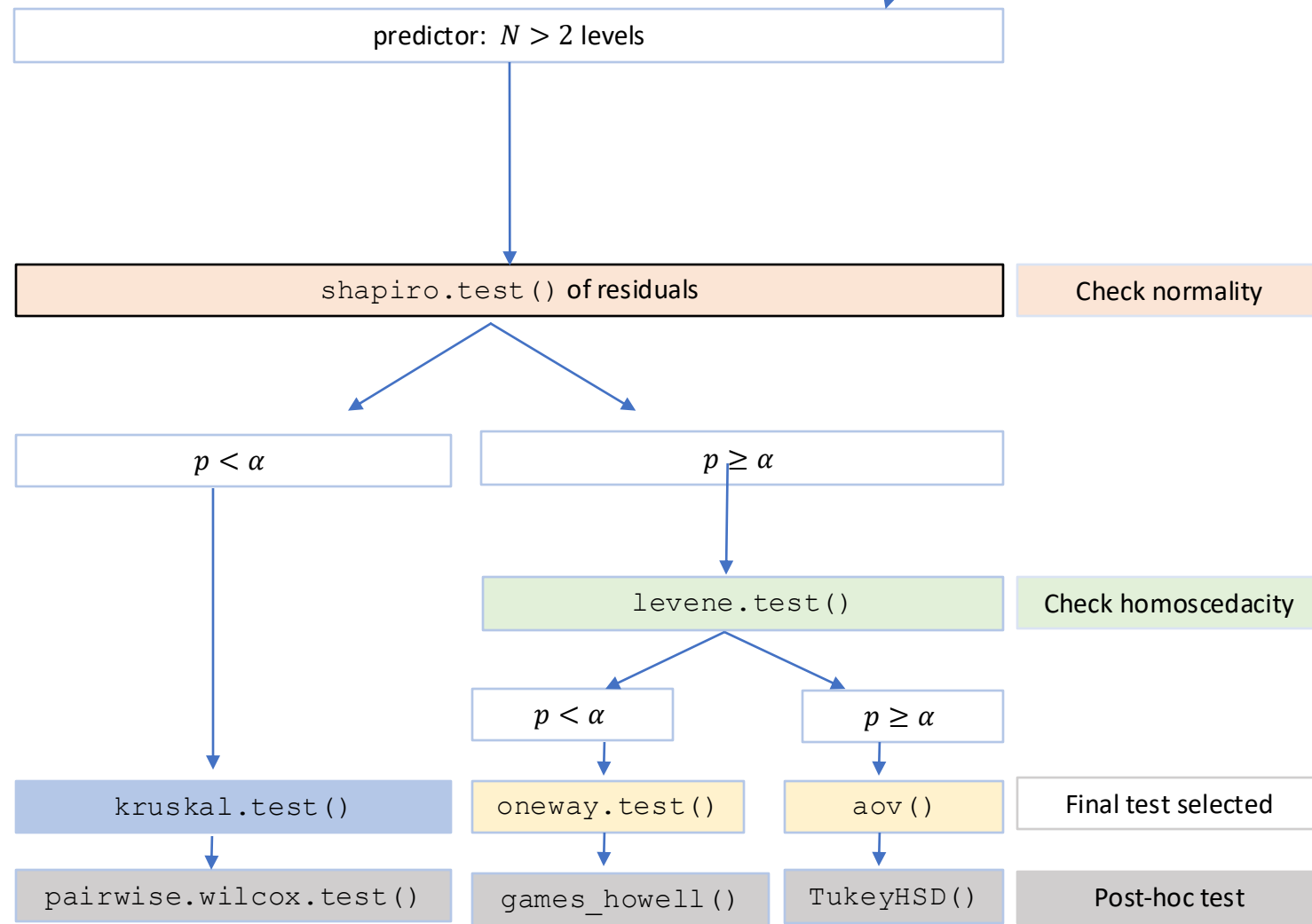
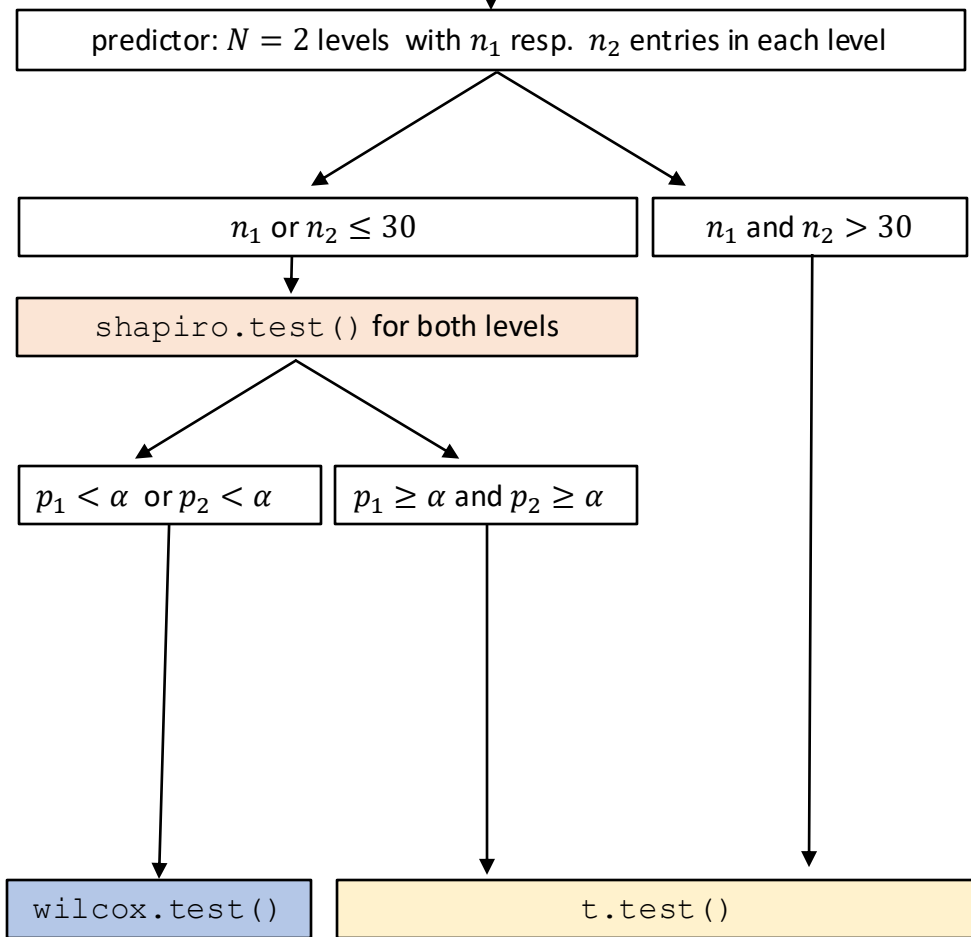


- Categorical predictor of class "factor" with N levels with n_i entries in each level, $i = 1, 2, \dots, N$
- Numeric response of class "numeric" or "integer"



- Categorical predictor of class “factor” with N levels
- Numeric response of class “numeric” or “integer”



Colour code:

parametric tests

non-parametric tests

predictor with $N = 2$ levels

predictor with $N > 2$ levels

$\alpha = 1 - \text{conf.level}$
 $p_i = p\text{-value of level } i, i = 1, 2$

- Categorical predictor of class "factor" with N levels
- Numeric response of class "numeric" or "integer"

`shapiro.test()` of residuals

Check normality

$p < \alpha$

$p \geq \alpha$

`levene.test()`

Check homoscedacity

$p < \alpha$

$p \geq \alpha$

$N = 2$

$N > 2$

$N = 2$

$N > 2$

$N = 2$

$N > 2$

`wilcox.test()`

`kruskal.test()`

`pairwise.wilcox.test()`

`t.test()`

`oneway.test()`

`games_howell()`

`t.test(..., var.equal=TRUE)`

`aov()`

`TukeyHSD()`

Final test selected

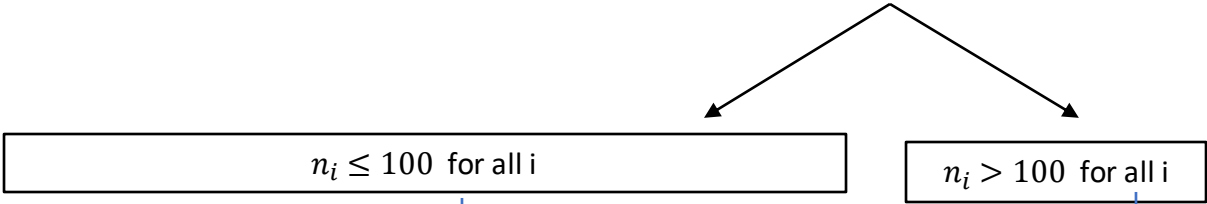
Post-hoc test

Colour code:

parametric tests	predictor with $N = 2$ levels
non-parametric tests	predictor with $N > 2$ levels

$\alpha = 1 - \text{conf.level}$
 $p_i = p\text{-value of level } i, i = 1, 2$

- Categorical predictor of class “factor” with N levels
- Numeric response of class “numeric” or “integer”



shapiro.test() of residuals

$p < \alpha$

$p \geq \alpha$

levene.test()

$p < \alpha$

$p \geq \alpha$

$N = 2$

$N > 2$

wilcox.test()

kruskal.test()

$N = 2$

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t.test()

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$N = 2$

$N > 2$

t.test(..., var.equal=TRUE)

aov()

pairwise.wilcox.test()

games_howell()

TukeyHSD()

Check normality

Check homoscedacity

Final test selected

Post-hoc test

Colour code:

parametric tests

predictor with $N = 2$ levels

non-parametric tests

predictor with $N > 2$ levels

$\alpha = 1 - \text{conf.level}$
 $p_i = p\text{-value of level } i, i = 1, 2$